AlphaGateau for Breakthrough: A piece of cake?

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Abstract

We study whether a board-size-invariant graph encoder improves self-play learning and cross-board transfer in Breakthrough. Using an AlphaZero-style training loop, we compare a graph neural network (GNN) with a convolutional neural network (CNN) on 8×8 play and in 5×5→8×8 transfer. Under our training budget, the CNN reaches a 100% greedy-evaluation win rate by iteration 40, while the GNN plateaus near 42%. Pretraining on 5×5 improves early 8×8 performance from 17% to 67% at iteration 5, but does not remove the CNN's long-run advantage. In a direct 12-game match, the CNN defeats the GNN 7 games to 5.

1 Introduction

Breakthrough is a two-player abstract strategy game played on an 8×8 board in which each side controls only pawns. Despite its simple rules, the game is strategically rich and has a large state space of approximately 1.5×10^{16} [2]. Figure 1 illustrates the starting position and the legal forward and diagonal-capture moves that define the game.

Breakthrough is a useful testbed for representation learning because strong play depends on local, directional structure and the same move rules apply across board sizes. We study whether an AlphaGateau-style graph neural network (GNN) can outperform a simpler convolutional neural network (CNN) on 8×8 Breakthrough, and whether its board-size-invariant representation makes 5×5 → 8×8 transfer meaningful. Under our training budget, the CNN is stronger from scratch, while 5×5 pretraining gives the GNN a clear early transfer benefit without closing the full gap.

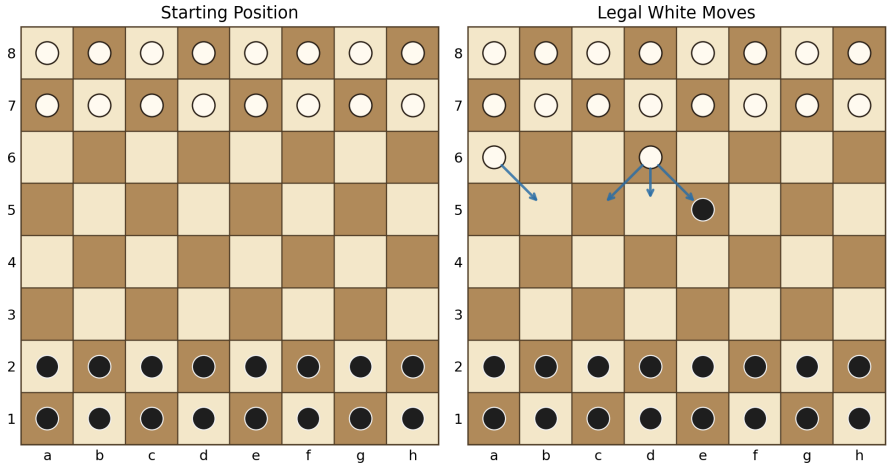

Figure 1: Breakthrough rules overview. *Left*: the standard starting position on an 8×8 board. *Right*: example legal moves for White. A pawn may move one square straight forward into an empty square, or one square diagonally forward only when capturing an opposing pawn.

2 Background

AlphaZero-style agents combine self-play reinforcement learning with Monte Carlo Tree Search (MCTS) [5]. A neural network predicts a policy over actions and a scalar value estimate; MCTS then refines action selection during play, and the search distribution becomes a policy target for training. We use Gumbel MCTS,[1] which selects actions by differentiably perturbing the policy logits with Gumbel noise [1].

AlphaGateau [3] replaces the convolutional encoder in AlphaZero with a graph neural network, aiming to better represent the relational structure of board games. It uses graph attention [6] with residual message passing layers.

Prior work on Breakthrough has focused primarily on exact game solving rather than self-play learning [4, 2]. We defer those details to Appendix A because our focus is representation learning and transfer rather than solving methodology. Our setting is different: we do not attempt to solve Breakthrough exactly, but instead study whether learned policy/value models can play effectively under self-play and transfer across board sizes.

3 Methodology

Environment and action encoding. We implemented a custom Breakthrough environment in JAX. States include the board array, an observation tensor, a legal-action mask, the side to move, rewards, and terminal game state. Actions use a fixed encoding $\texttt{action} = \texttt{from_square} \times 3 + \texttt{move_type}$, where $\texttt{move_type} \in \{\texttt{forward}, \texttt{diag_left}, \texttt{diag_right}\}$. We canonicalize the board so the player to move always advances upward, keeping both the tensor and graph interfaces board-size agnostic.

Model representations. The CNN baseline uses a three-channel tensor (self pieces, opponent pieces, empty squares) fed into a residual convolutional network. The GNN constructs a graph with one node per square and directed move edges for both players.[2] Node features encode occupancy, normalized coordinates, and edge-of-board indicators; edge features encode move direction, owner, legality, capture structure, and goal-rank information. This graph construction preserves the same local feature schema across board sizes, which is the basis for our transfer experiments.

Learning architectures and training procedure. For the AlphaZero-style agents, both the CNN and GNN use hidden size 64 and four residual layers. The GNN adds attention pooling over node embeddings to produce a global representation. Self-play training uses 32 MCTS simulations per move, a 50k-sample replay buffer, and 32-sample minibatches. Unless otherwise noted, the learning rate is 10^{-3}.

Experiment configurations. We ran three main experiment groups. First, in the 8×8 scratch comparison, the CNN and GNN were trained from scratch for 40 iterations[3] with 64 self-play games per iteration and a 192-ply move cap. Second, in the transfer setup, we pretrained the GNN on 5×5 for 40 iterations using 128 self-play games per iteration, then fine-tuned it on 8×8 for 30 iterations.[4] Third, we repeated selected GNN runs with cosine decay and explored a progressive $5\times5 \rightarrow 6\times6 \rightarrow 8\times8$ transfer ablation. We also evaluated a value-only TD(λ) baseline, deferred to Appendix B.

Evaluation protocol. Every five iterations we play 12 games against a greedy evaluator and record the win rate. For the policy-based agents, the greedy evaluator always picks the highest-policy action. After training, we run a 12-game head-to-head match between the final GNN and CNN scratch checkpoints. We also record zero-shot 8×8 performance of the 5×5 pretrained model before any fine-tuning. For the progressive transfer experiment, we evaluate each stage only on its current

[1] using the `mctx` library

[2] This is in line with the AlphaGateau paper.

[3] We used 40 iterations as the standard budget because of our single-GPU compute constraints.

[4] We used 30 fine-tuning iterations to measure early 8×8 transfer gains rather than fully optimize the transferred model.

Run	Iters	Peak win rate	Final win rate
GNN 8×8 scratch	40	0.67 @ iter 20	0.42 @ iter 40
CNN 8×8 scratch	40	1.00 @ iter 40	1.00 @ iter 40
GNN 5×5 pretrain	40	1.00 @ iter 20	1.00 @ iter 40
GNN 8×8 fine-tune	30	0.75 @ iter 15	0.67 @ iter 30
GNN 5×5 pretrain (cosine decay)	40	1.00 @ iter 20	0.83 @ iter 40
GNN 8×8 scratch (cosine decay)	40	0.58 @ iter 40	0.58 @ iter 40
GNN 8×8 fine-tune (cosine decay)	30	0.83 @ iter 15	0.83 @ iter 30
GNN 6×6 fine-tune (cosine, progressive)	30	1.00 @ iter 20	0.92 @ iter 30
GNN 8×8 fine-tune (cosine, progressive)	30	0.92 @ iter 5	0.58 @ iter 30

Table 1: Win rates against the greedy evaluator over 12 games. Cosine and progressive-transfer runs are included because they change the transfer outcome.

Final GNN variant	Wins	Losses	Win rate
Scratch GNN	2	10	0.17
Scratch GNN (cosine decay)	1	11	0.08
Direct 5×5→8×8 transfer (cosine)	4	8	0.33
Progressive 5×5→6×6→8×8 transfer (cosine)	5	7	0.42

Table 2: Direct 12-game matches against the final CNN 8×8 scratch checkpoint. Despite the small sample, they show how much each GNN variant narrows the final gap.

board size; we do not re-evaluate the final 8×8 checkpoint on earlier boards, so these results reflect stagewise transfer rather than catastrophic forgetting.

4 Results

Table 1 summarises the main training runs, Table 2 reports the direct 12-game matches, and Figure 2 shows the 8×8 scratch comparison.

Scratch comparison. The CNN and GNN diverge sharply after iteration 15. By iteration 40, the CNN reaches 1.00 greedy-evaluation win rate while the scratch GNN finishes at 0.42 (Table 1), and the CNN wins the final head-to-head 10 games to 2 (Table 2). Figure 2 shows the same pattern: the CNN improves steadily while the GNN peaks earlier and plateaus.

Transfer learning. The 5×5 pretrained GNN converges cleanly, reaching 1.00 win rate by iteration 20 and maintaining it through iteration 40. Applied zero-shot to the 8×8 board, it achieves only 17% win rate (2/12 games), showing that the 5×5 policy does not transfer directly to the larger board.

Fine-tuning from the 5×5 checkpoint gives a substantial early-stage advantage. At iteration 5, the GNN pretrained on 5×5 and then fine-tuned directly on 8×8 with cosine decay wins 75% of games versus a greedy evaluator, compared to 17% for the fixed-LR GNN trained from scratch at the same iteration.[5] This shows meaningful cross-board transfer. By iteration 30, this direct 5×5 → 8×8 fine-tuning run reaches 83%, while the fixed-LR scratch GNN at the same iteration is at 58%, a persistent advantage. This is the strongest transfer run in Table 1 by final greedy-evaluator win rate, so we use it as the main transfer result. The final transfer checkpoint still trails the final CNN checkpoint overall, but in direct final-checkpoint play it narrows the gap from 2 wins in 12 games for the scratch GNN to 4 wins in 12 games against the same CNN (Table 2). A progressive 5×5 → 6×6 → 8×8 pipeline does not improve the final 8×8 greedy-evaluator result, so we treat it as a secondary ablation and defer its full curves to Appendix C.

[5] more than a 4× improvement

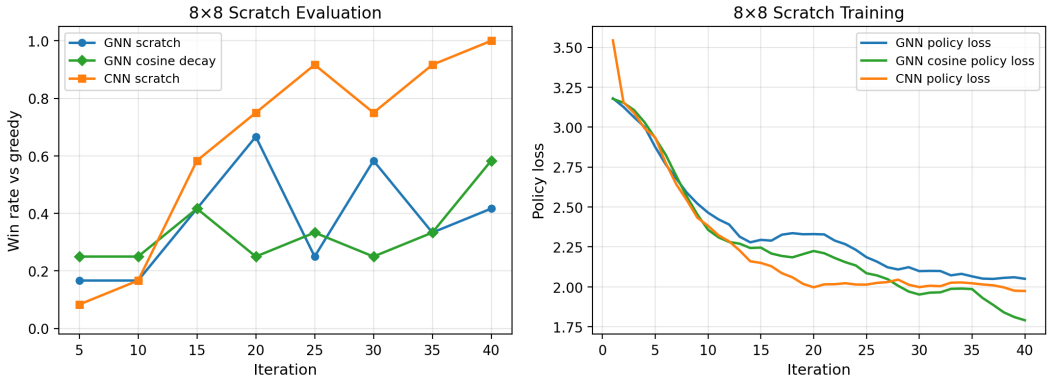

Figure 2: 8×8 scratch comparison. *Left*: greedy-evaluation win rate over training iterations; the CNN (orange) overtakes the GNN (blue) at iteration 15 and reaches 1.00 by iteration 40 while the GNN oscillates and plateaus. *Right*: policy loss curves confirming the GNN's convergence failure.

5 Discussion

Why the CNN outperforms the GNN. The CNN's advantage on 8×8 Breakthrough is a meaningful negative result. On a regular grid, convolutions share weights across spatial positions and efficiently capture the local, directional patterns that dominate Breakthrough tactics. The GNN adds extra optimization burden through attention pooling and edge-feature processing, which appears harder to train under the fixed 40-iteration budget. Figure 2 is consistent with this interpretation.

Transfer benefit and limits. Transfer from 5×5 pretraining produces a real, quantifiable early-training benefit: the fine-tuned model starts 4× higher than the scratch GNN at iteration 5. This confirms our second hypothesis: the GNN's board-size-invariant representation supports meaningful knowledge transfer. However, the benefit is largest early in fine-tuning and shrinks as both models train longer, suggesting that pretraining provides a stronger initialisation rather than a compounding structural advantage. Using cosine decay improves the final transferred checkpoint to 0.83 and reduces the late-stage drop seen with a fixed learning rate.

6 Conclusion and Future Work

We adapted AlphaGateau to Breakthrough and found that, under the current 40-iteration budget, the CNN remains the strongest model on 8×8 from scratch. GNN transfer from 5×5 meaningfully improves early 8×8 learning and narrows the final head-to-head gap to the CNN. Our main transfer result is the GNN pretrained on 5×5 and then fine-tuned directly on 8×8 with cosine decay, because it achieves the best final greedy-evaluator score among the transfer runs. The most useful next steps are to extend the fine-tuning budget, increase the evaluation game count, and run GNN ablations that simplify attention pooling or edge features.

7 Contributions and Acknowledgements

Contributions. Athmane Benarous focused on the JAX environment, training pipeline, Slurm orchestration, code pipeline setup, CNN model training, report drafting, and head-to-head comparisons. Farooq Khan focused on the transfer-learning baseline, head-to-head comparisons, presentation slides, video recording, and report writing. Andrew Saffar focused on transfer learning, the TD(λ) baseline, presentation slides, video recording, and report drafting.

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A Additional Background

Prior work on Breakthrough has focused primarily on exact game solving rather than self-play learning. Early work combined race patterns with job-level proof-number search to solve the 6×5 variant as a second-player win [4]. More recently, the 6×6 game was weakly solved as a first-player win using a hybrid of endgame tablebases, race patterns, proof-number-based search, and a neural-network-guided job scheduler [2]. These results show that, despite simple rules, Breakthrough is strategically rich enough to support substantial search and solving research.

B TD(λ) Baseline

For TD(λ), we use $\lambda = 0.7$ and ϵ-greedy self-play; the 5×5 run uses 128 episodes per iteration and a 96-ply move cap, while the 8×8 run uses 64 episodes per iteration and a 256-ply move cap, with both running for 40 iterations at learning rate 10^{-3}.

Evaluation protocol. Since the TD baseline has no policy head, it is evaluated using greedy value-based play, selecting the action whose successor state has the highest predicted value rather than using MCTS.

TD(λ) baseline Results. TD(λ) remains much weaker than the MCTS-based agents against the greedy evaluator. On the 5×5 board it reaches a peak win rate of 1.00 at iteration 10 before falling back to 0.83 by iteration 40, and on the 8×8 board it only reaches 0.58 by iteration 40. This profile is consistent with a value learner that can exploit some short-horizon patterns on the smaller board but does not produce a policy strong enough to compete reliably on the full game.

TD(λ) baseline Discussion. Even when TD(λ) wins some evaluation games, it still trails the search-based models by a wide margin. We attribute this to the evaluator's explicit lookahead: TD(λ) must encode that lookahead implicitly in its value function, while the AlphaZero-style agents obtain it directly through MCTS. Under the current compute budget, the results therefore support the need for search-guided policy learning rather than value-only self-play.

C Additional Training Curves

Progressive transfer curves. Figure 3 shows the full 5×5 → 6×6 → 8×8 pipeline. Transfer to the intermediate 6×6 task is strong, but the benefit does not carry cleanly to the final 8×8 stage, which finishes below the GNN pretrained on 5×5 and then fine-tuned directly on 8×8 with cosine decay. Because we do not re-evaluate the final checkpoint on earlier board sizes, this figure should be read as stagewise transfer evidence rather than a direct measurement of catastrophic forgetting.

D Encoding Figure

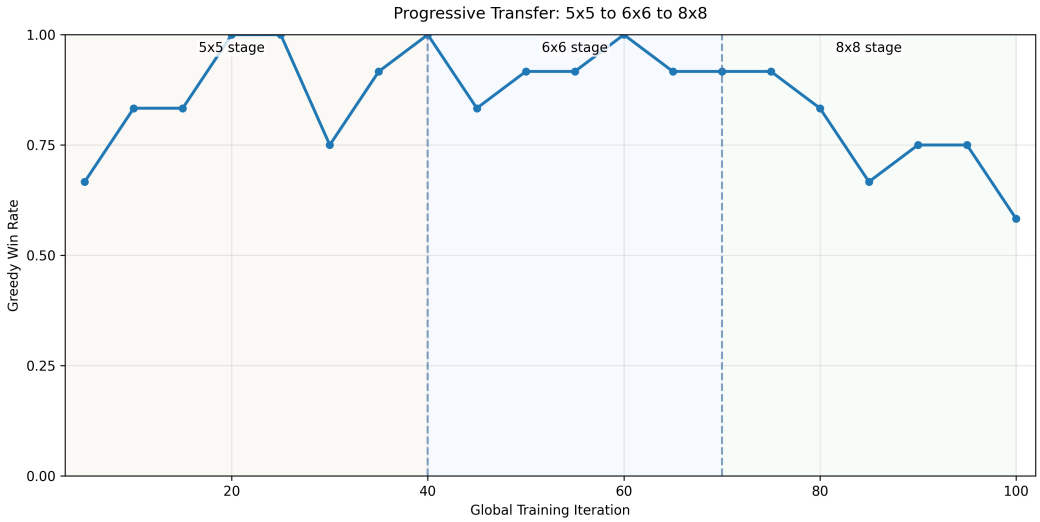

Figure 3: Progressive transfer pipeline across board sizes. Vertical dashed lines mark stage boundaries and shaded bands indicate the active board size. The model adapts well to 6×6 but does not retain that advantage through the final 8×8 fine-tuning stage. Each win-rate point is measured over 12 greedy-evaluation games.

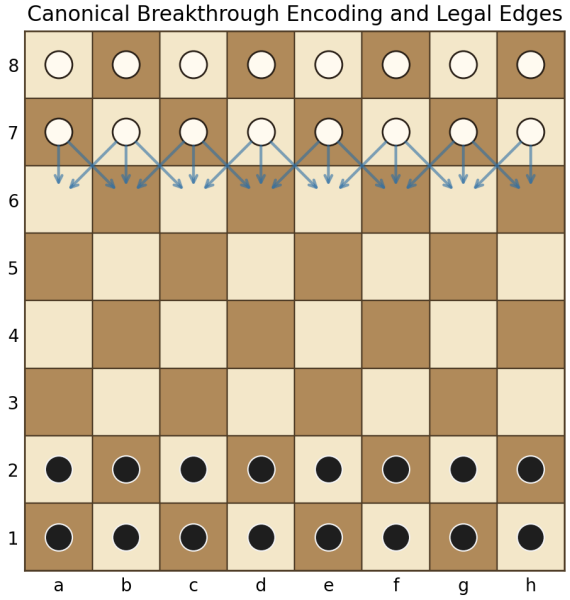

Figure 4: Canonical Breakthrough encoding showing legal directed edges for the current player at game start. Each square is a graph node; directed arrows indicate legal forward and diagonal captures.